

Maharashtra Public Service Commission Assessment

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Assistant Professor and Associate Professor Electronics and Tele Communication Group A

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Assistant Professor and Associate Professor Electronics and Tele Communication Group A

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Question Number : 1 Question Id : 464662101 Question Type : MCQ Option Shuffling : No

If 'a' is a constant vector and $r = |\vec{r}|$ then $\text{a} \cdot \nabla \left(\frac{1}{r} \right)$ is (where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$)

Options :

1. $-\frac{\text{a} \cdot \vec{r}}{r^2}$

2. $\frac{\text{a} \cdot \vec{r}}{r^2}$

3. $-\frac{\text{a} \cdot \vec{r}}{r^3}$

4. $\frac{\text{a} \cdot \vec{r}}{r^3}$

The system of linear simultaneous equations

$$3x - y + 4z = 3$$

$$x + 2y - 3z = -2$$

$$6x + 5y + \lambda z = -3$$

have unique solution only if

Options :

1. $\lambda \neq -3$

2. $\lambda \neq -5$

3. $\lambda \neq -2$

4. $\lambda \neq 1$

Question Number : 3 Question Id : 464662103 Question Type : MCQ Option Shuffling : No

If $A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 3 & 6 & 9 & 3 \\ -2 & 4 & 2 & 6 \\ 4 & 8 & 12 & 4 \end{bmatrix}$ then the rank of matrix A is

Options :

1. 1

2. 2

3. 3

4. 4

Solution of initial value problem $x'' + x' + 2x = 0$; $x(0) = 0$; $x'(0) = 2$ is

Options :

1.
$$\phi(t) = \frac{7}{2} \left(\sin \frac{7}{2} t \right) e^{-t/2}$$

2.
$$\phi(t) = \frac{7}{4} \left(\cos \frac{7}{2} t \right) e^{-t/2}$$

3.
$$\phi(t) = \left(\cos \frac{5}{2} t + \sin \frac{5}{2} t \right) e^{+(3/2)t}$$

4.
$$\phi(t) = \frac{4}{7} \left(\sin \frac{7}{2} t \right) e^{-t/2}$$

Question Number : 5 Question Id : 464662105 Question Type : MCQ Option Shuffling : No

For what value of 'a', the integral $\int_0^{\infty} e^{-ax} dx$ is convergent ?

Options :

1. $a > 0$

2. $a \geq 0$

3. $a < 0$

4. For all values of a

Question Number : 6 Question Id : 464662106 Question Type : MCQ Option Shuffling : No

The differential equation $(4x + 3y + 1)dx + (3x + 2y + 1)dy = 0$ represents a family of

Options :

1. Hyperbola
2. General equation of circle
3. Parabola
4. Ellipse

If $f(x, y, z) = e^{x+y+z}$, then the direction of maximum rate of change of f , at the point $(1, 2, 3)$ is along the vector

Options :

1. $\bar{i} + \bar{j} + \bar{k}$

2. $\bar{i} + 2\bar{j} + 3\bar{k}$

3. $e^6(\bar{i} + \bar{j} + \bar{k})$

4. $e(\bar{i} + 2\bar{j} + 3\bar{k})$

Question Number : 8 Question Id : 464662108 Question Type : MCQ Option Shuffling : No

Evaluate : $\int_C \frac{e^{2z} dz}{(z+1)^4}$ where C is $|z| = 3$.

Options :

1. $\frac{8\pi i}{3e^2}$

2. $\frac{4\pi i}{5e^2}$

3. $\frac{3\pi i}{7e^4}$

4. $\frac{5\pi}{3}$

Question Number : 9 Question Id : 464662109 Question Type : MCQ Option Shuffling : No

Let $f(z) = u + iv$ be analytic function whose real part is $u = x^3 - 3xy^2 + 3x^2 - 3y^2$.
Then the imaginary part of $f(z)$ is given by

Options :

1. $3xy^2 + xy - y^3 + c$

2. $3x^2y - y + 1 + c$

3. $3x^2y + 6xy - y^3 + c$

4. $2x^2y + 6xy - 1$

Question Number : 10 Question Id : 464662110 Question Type : MCQ Option Shuffling : No

The correct relationship between A.M., G.M. and H.M. is

Options :

1. $A.M. = G.M. = H.M.$

2. $G.M. \geq A.M. \geq H.M.$

3. $H.M. \geq G.M. \geq A.M.$

4. $A.M. \geq G.M. \geq H.M.$

A coin is tossed 4 times. What is the probability of getting heads exactly 3 times ?

Options :

1. $\frac{1}{4}$

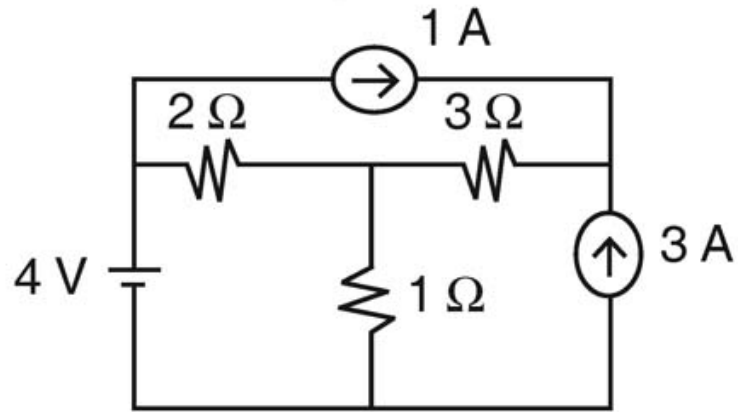
2. $\frac{3}{8}$

3. $\frac{1}{2}$

4. $\frac{3}{4}$

Question Number : 12 Question Id : 464662112 Question Type : MCQ Option Shuffling : No

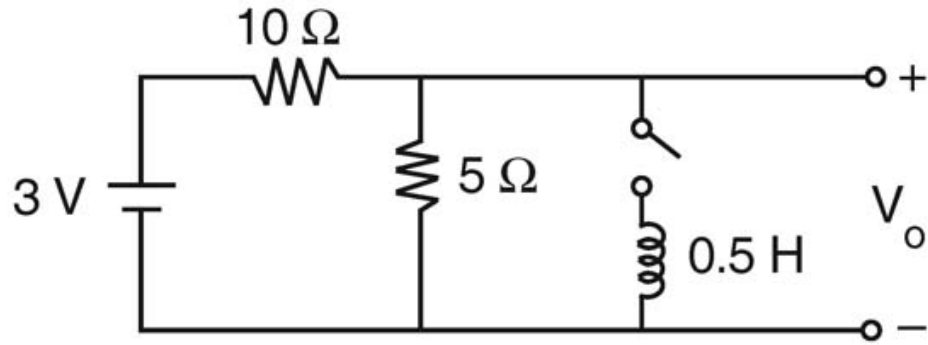
In the following circuit the current through $1\ \Omega$ resistor is



Options :

1. 2 amp
2. 4 amp
3. 1 amp
4. 5 amp

In the network shown in following figure a steady-state condition is achieved with switch open. At $t = 0$ switch is closed.



The output voltage $V_o(t)$ is

Options :

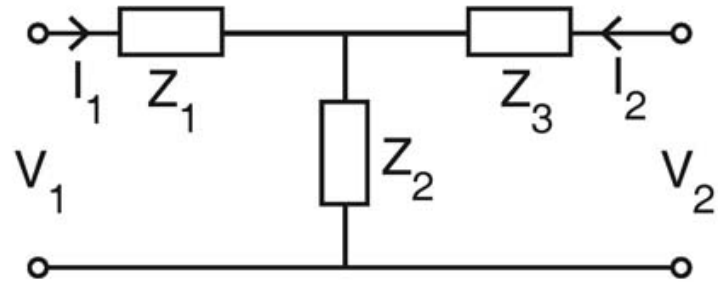
1. $V_o(t) = e^{-20/3t} V$

2. $V_o(t) = e^{-10/3t} V$

3. $V_o(t) = e^{-3/10t} V$

4. $V_o(t) = e^{-3/20t} V$

Z-parameters of a two port network are $Z_{11} = 10 \Omega$, $Z_{12} = Z_{21} = 5 \Omega$, $Z_{22} = 20 \Omega$. Find the equivalent T network.



Options :

1. $Z_1 = 5, Z_2 = 10, Z_3 = 15$
2. $Z_1 = 5, Z_2 = 5, Z_3 = 10$
3. $Z_1 = 5, Z_2 = 5, Z_3 = 5$
4. $Z_1 = 5, Z_2 = 5, Z_3 = 15$

For the following signal

$$X[n] = \begin{cases} 1, & |n| \leq 2 \\ 0, & \text{otherwise.} \end{cases}$$

The DFT of above signal is

Options :

1. $\frac{\sin 5 \Omega}{\sin \Omega}$

2. $\frac{\sin 4 \Omega}{\sin \Omega}$

3. $\frac{\sin 2.5 \Omega}{\sin 0.5 \Omega}$

4. None of these

Find the initial and final values of $X(s) = \frac{8s + 12}{s(s + 3)}$ using initial and final value theorem.

Options :

1. 8, 3

2. 3, 8

3. 8, 4

4. 3, 4

Which two port network parameter set directly relates input and output voltages and currents using admittance values ?

Options :

1. Y-parameters
2. H-parameters
3. T-parameters
4. Z-parameters

A LTI system with its impulse response given by $h(n) = \delta(n)$. What will be its response to unit step sequence ?

Options :

1. $y(n) = u(n)$
2. $y(n) = u(n - 1)$
3. $y(n) = -u(-n - 1)$
4. $y(n) = -u(n)$

The linear convolution of two sequences given by

$$x_1(n) = u(n) - u(n - 3)$$

and $x_2(n) = u(n) - u(n - 2)$ is

Options :

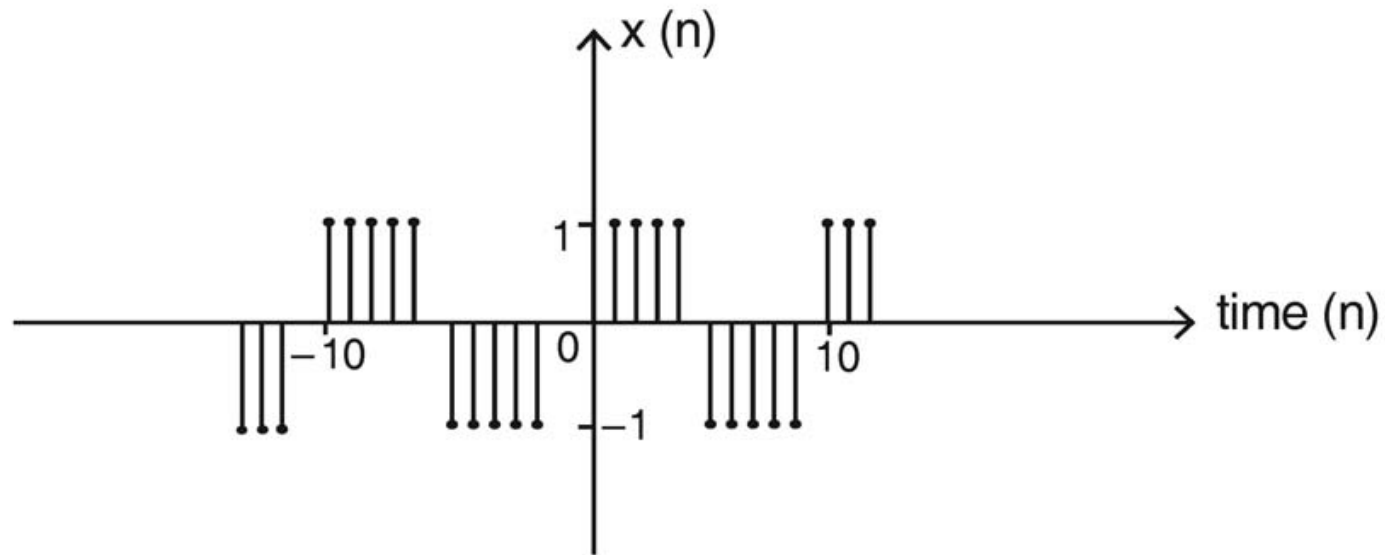
1. $x_3(n) = \{1, 2, 2, 1\}$

2. $x_3(n) = u^2(n) - u(n - 2) - 4(n - 3) + u(n - 2) \cdot u(n - 3)$

3. $x_3(n) = u(n) - u(n - 4)$

4. $x_3(n) = \{1, 1, 1, 1, 1\}$

The fundamental frequency of discrete-time square-wave signal shown in fig. is



Options :

1. $\frac{\pi}{10}$ rad/sec
2. $\frac{\pi}{5}$ rad/sec
3. 10π rad/sec
4. 10 rad/sec

Question Number : 21 Question Id : 464662121 Question Type : MCQ Option Shuffling : No

z-transform of a sequence $x(n)$ is given as $X(z) = 3 + 2z - \frac{3}{z} + \frac{5}{z^2}$. If $y(n)$ is the first difference of $x(n)$, then $Y(z)$ is given by

Options :

1. $1 + 2z - \frac{6}{z} + \frac{8}{z^2} - \frac{5}{z^3}$

2. $-1 - 2z - \frac{6}{z} + \frac{8}{z^2} + \frac{5}{z^3}$

3. $-3z + 2z^2 - \frac{3}{z^2} + \frac{5}{z^3}$

4. $3z + 2z^2 - \frac{3}{z^2} + \frac{5}{z^3}$

Question Number : 22 Question Id : 464662122 Question Type : MCQ Option Shuffling : No

The inverse Laplace transform of $F(s) = \frac{1}{s(s+1)}$ is

Options :

1. $K e^{-t}$

2.

3. $1 - e^{-t}$

4.

5. $e^{-t} - 1$

6.

7. $1 + e^{-t}$

8.

Question Number : 23 Question Id : 464662123 Question Type : MCQ Option Shuffling : No

Fourier transform of $x(t) = e^{2t} \cdot u(t)$

Options :

1. does not exist but LT exists for $\sigma > 2$
2. does exist and LT exists for $\sigma > 2$
3. does not exist but LT exists for $\sigma < 1$
4. does not exist and LT also does not exist

Question Number : 24 Question Id : 464662124 Question Type : MCQ Option Shuffling : No

If the Laplace transform of the $y(t)$ is $Y(s) = \frac{1}{s(s-1)}$ then the final value is

Options :

1. -1

2. 0

3. 1

4. None of these

In an intrinsic semiconductor at temperature of absolute 0° Kelvin, which of the following statement is correct ?

Options :

1. All energy states in the valence band are empty of electrons and all energy states in conduction band are filled with electrons.
2. All energy states in the valence band are filled with electrons and all energy states in conduction band are empty of electrons.
3. All energy states in the valence and conduction band are filled with holes.
4. All energy states in the valence and conduction band are filled with electrons.

Question Number : 26 Question Id : 464662126 Question Type : MCQ Option Shuffling : No

The process of doping in intrinsic semiconductor

Options :

1. Does not change conductivity
2. Increases conductivity
3. Decreases conductivity
4. None of the above

Question Number : 27 Question Id : 464662127 Question Type : MCQ Option Shuffling : No

A semiconductor wafer is 0.5 mm thick. A potential of 100 mV is applied across the thickness. What is the electron drift velocity if their mobility is $0.2 \text{ m}^2/\text{V.s}$?

Options :

1. 40 m/s
2. 20 m/s
3. 0.4 m/s
4. 0.2 m/s

Question Number : 28 Question Id : 464662128 Question Type : MCQ Option Shuffling : No

In a PN junction diode, under reverse bias condition, the magnitude of the electric field is maximum at

Options :

1. The edge of the depletion region on the n-side
2. The edge of the depletion region on the p-side
3. The physical PN junction
4. The centre of the depletion region on the n-side

Photodiode operates in two modes namely

- Current mode
- Voltage mode

Which of the following statement is correct ?

Options :

1. The photodiode operated in both current mode and voltage mode, needs external voltage source
2. The photodiode operated in current mode needs external voltage source, but in the voltage mode, it does not require any external voltage source
3. The photodiode operated in current mode does not require external voltage source, but in the voltage mode requires external voltage source
4. The photodiode operated in current mode and voltage mode does not require any external voltage source

In an intrinsic semiconductor at thermal equilibrium, which of the following is correct ?

Options :

1. Number of electrons is greater than holes
2. Number of holes is greater than electrons
3. Number of electrons equals number of holes
4. No charge carriers exist

“Early Effect” in a bipolar junction transistor refers to a reduction of the effective base width caused by

Options :

1. Forward biasing of emitter base junction
2. The electron hole recombination at the base
3. The reverse biasing of the base-collector junction
4. The early removal of stored base charge during saturation to cut-off switching

Question Number : 32 Question Id : 464662132 Question Type : MCQ Option Shuffling : No

In a pure Si bar of length $1\text{ }\mu\text{m}$ long and $100\text{ }\mu\text{m}^2$ cross-sectional area, assume that $n \approx 10^{18}\text{ cm}^{-3}$ and the velocity of electrons is saturated at 10^7 cm/s . Find time to drift $1\text{ }\mu\text{m}$ for an electric field of 10^6 V/cm .

Options :

1. 1 ps
2. 10 ps
3. 100 ps
4. $1\text{ }\mu\text{s}$

MOSFET can be used as

Options :

1. Current controlled capacitor
2. Current controlled inductor
3. Voltage controlled capacitor
4. Voltage controlled inductor

Which of the following statements is true for a forward biased pn-junction ?

Options :

1. The forward current is primarily due to the diffusion of minority carriers in the other side of the junction
2. The forward current is primarily due to the drift of the majority carriers to the other side of the junction
3. The forward current share of drift and diffusion currents is almost equal
4. None of the above

Question Number : 35 Question Id : 464662135 Question Type : MCQ Option Shuffling : No

Following values are given for a p^+n -junction diode : $V_T \approx 26 \text{ mV}$, $N_a \approx 10^{17} \text{ cm}^{-3}$, $N_d \approx 10^{13} \text{ cm}^{-3}$, $n_i \approx 10^{10} \text{ cm}^{-3}$. Assuming $\ln(10) = 2.3$, calculate the barrier potential V_{bi} for the diode.

Options :

1. 0.394 V
2. 0.598 V
3. 1.327 V
4. 2.578 V

For a p-type MOS capacitor, strong inversion occurs when the surface potential equals

Options :

1. ϕ_F
2. $2\phi_F$
3. $3\phi_F$
4. $\phi_{F/2}$

The invention of a blue LED was very significant because

Options :

1. Blue colour is the most widely used in LEDs.
2. It was possible to get white light by using blue LEDs.
3. Blue laser is relatively harmless as compared to other lasers.
4. It was invented before red and green LEDs.

Larger the value of Filter capacitor

Options :

1. Larger the value of peak to peak ripple voltage
2. Larger the peak current in the rectifying diode
3. Longer the time that current pulse flow through the diode
4. Smaller the dc voltage across the load

With reference to diode rectifier circuits with resistive load

- a) Half-wave rectifier has PIV of V_m , ripple factor 1.21 and ripple frequency equal to f
- b) Full wave bridge rectifier has PIV of V_m , ripple factor 0.812 and uses 4 diodes
- c) Full wave center-tapped rectifier has PIV of $2 V_m$, ripple factor 0.482 and ripple frequency equal to $2f$
- d) Transformer utilization factor of Half-wave rectifier is 0.287, full wave center-tapped rectifier is 0.693 and full-wave bridge rectifier is 0.482

Which of the statement given above is/are correct ?

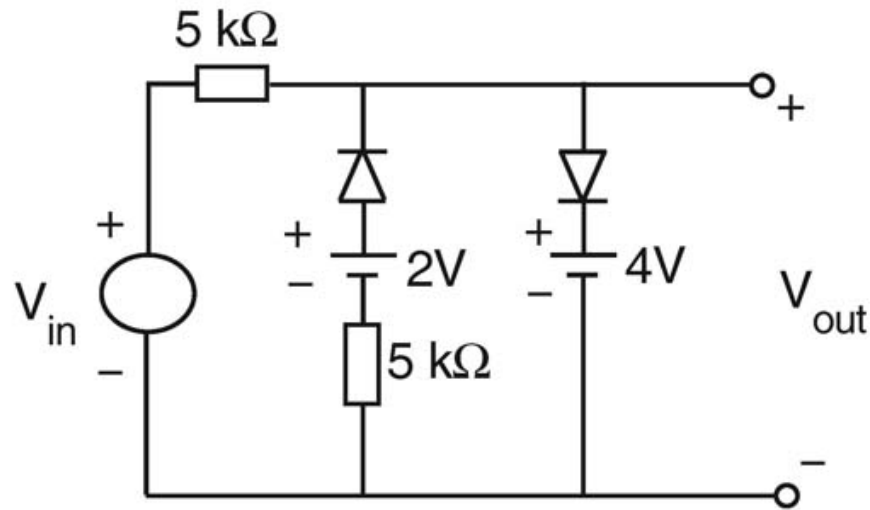
Options :

- 1. c only
- 2. a and c only
- 3. a and d only

b, c and d

4.

A voltage signal $10 \sin \omega t$ is applied to the circuit with ideal diodes, as shown in the figure. The Maximum and the minimum values of the output voltage V_{out} are, respectively (assuming $R_F = 0$ and $V_F = 0$),



Options :

1. $+4\text{ V}, -4\text{ V}$
2. $+10\text{ V}, -10\text{ V}$
3. $+4\text{ V}, -2\text{ V}$

4. +2 V, -2 V

Question Number : 41 Question Id : 464662141 Question Type : MCQ Option Shuffling : No

During off time of diode in a clamper circuit capacitor

Options :

1. Starts charge rapidly

2. Discharge through diode

3. Holds constant charge

4. None of the above

For two MOSFETs M_1 operates at low V_{ov} and M_2 operates at high V_{ov} . If both carry same I_D , which has higher transconductance ?

Options :

1. M_1

2. M_2

3. Both equal

4. Cannot determine

In common-emitter hybrid parameters, the ratio of change in the collector current to the corresponding change in the collector voltage with the base current kept constant is known as

Options :

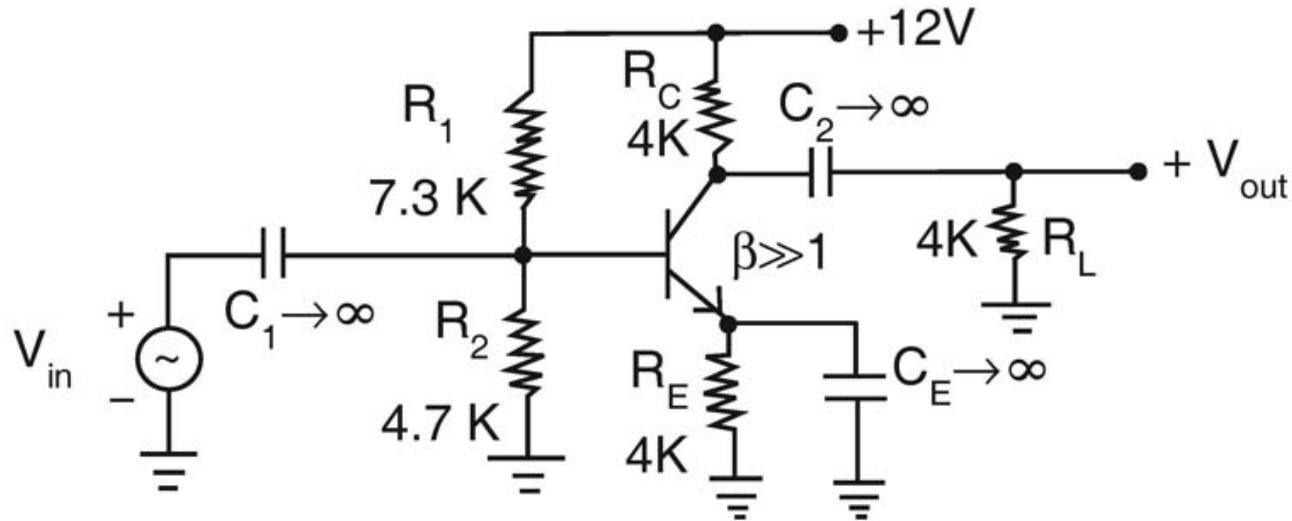
1. Output Admittance
2. Input Impedance
3. Reverse Voltage Gain
4. Output Impedance

Which parameter does not affect because of frequency scaling in active filter ?

Options :

1. Cutoff frequency
2. Bandwidth
3. Quality factor (Q)
4. RC time constant

Find approximate voltage gain of the circuit shown below. Given : $I_{CQ} = 1\text{mA}$, $V_{BE \text{ active}} = 0.7\text{ V}$, $V_T = 25\text{ mV}$ and all internal capacitors are negligible.



Options :

1. -40
2. -80
3. -160
4. -200

The average of the currents entering into the inverting input and non-inverting input terminals of an op-amp is called

Options :

1. Output bias current
2. Input offset current
3. Input bias current
4. Output offset current

Table gives properties of differential amplifier circuit configuration. The correct match is :

Configuration	Properties : Voltage gain and input resistance
a. Single Input Unbalanced Output	i. $A_d = \frac{R_c}{r_e}, R_{i1} = R_{i2} = 2\beta_{ac}R_E$
b. Dual input balanced output	ii. $A_d = \frac{R_c}{2r_e}, R_{i1} = R_{i2} = 2\beta_{ac}R_E$
c. Single input balanced output	iii. $A_d = \frac{R_c}{r_e}, R_i = 2\beta_{ac}R_E$
d. Dual input unbalanced output	iv. $A_d = \frac{R_c}{2r_e}, R_i = 2\beta_{ac}R_E$

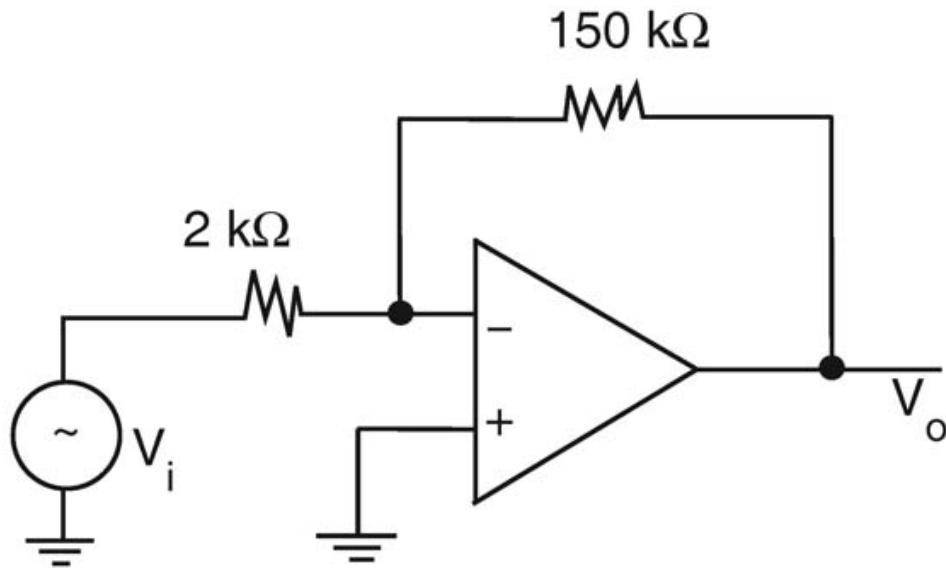
Options :

1. a - i, b - iii, c - iv, d - ii

2. a - iv, b - i, c - iii, d - ii

3. a - ii, b - iv, c - i, d - iii

4. a - iii, b - i, c - ii, d - iv

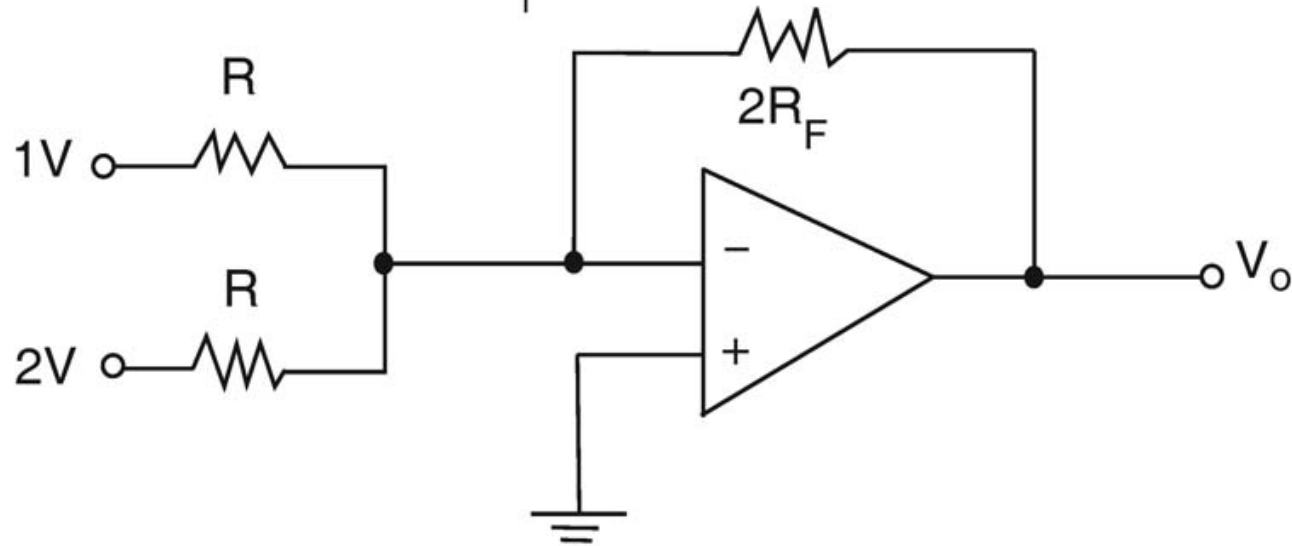


Calculate the output offset voltage of the Op-amp for the circuit shown. The Op-amp has input offset voltage $V_{io} = 1.2\text{mV}$.

Options :

1. 90 mV
2. Almost same
3. 91.2 mV
4. 120 mV

In order to have the output $V_o = -3$ V from an inverting summer shown in figure, the relation between R_f and R is



Options :

1. $\frac{R_f}{R} = 2$

2. $R_f = R$

3. $\frac{R_f}{R} = \frac{1}{2}$

4.
$$R_f = \frac{1}{R}$$

Question Number : 50 Question Id : 464662150 Question Type : MCQ Option Shuffling : No

An op-amp integrator uses

Options :

1. Resistor in feedback

2. Capacitor in feedback

3. Inductor in feedback

4. Diode in feedback

Question Number : 51 Question Id : 464662151 Question Type : MCQ Option Shuffling : No

$(-17)_{10}$ in one's complement form is

Options :

1. 010001

2. 101110

3. 110001

4. 110000

Question Number : 52 Question Id : 464662152 Question Type : MCQ Option Shuffling : No

A few addressing modes and a fixed instruction size are the features of

Options :

1. CISC
2. RISC
3. RAID
4. None of the above

Question Number : 53 Question Id : 464662153 Question Type : MCQ Option Shuffling : No

$\overline{\overline{AB} + \overline{AB} + A\overline{B}}$ is equivalent to

Options :

1. $A + B$

2. AB

3. $\overline{A} + \overline{B}$

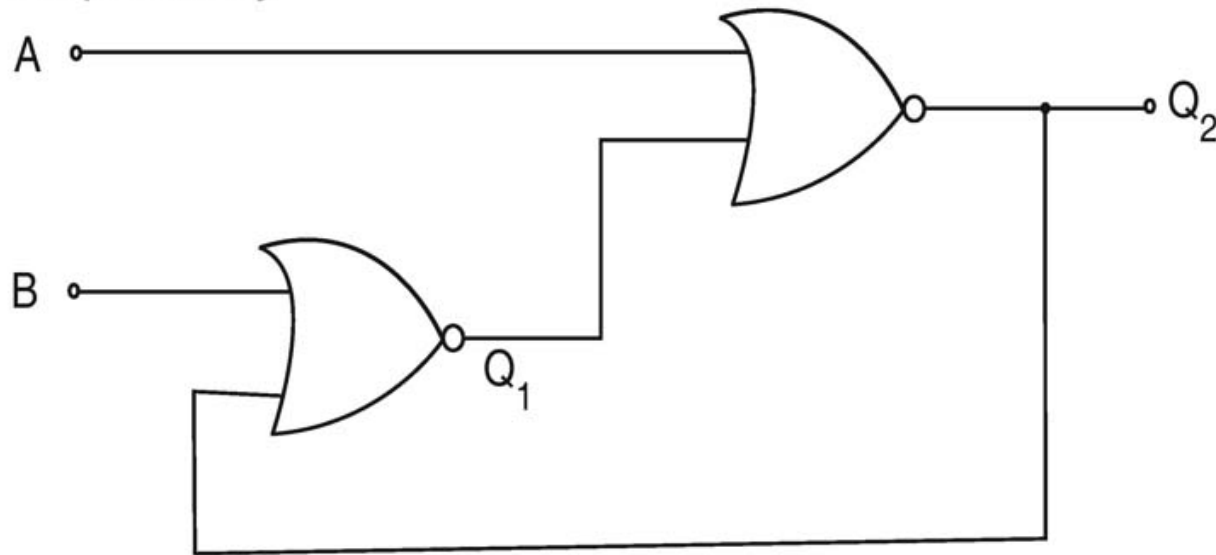
4. $\overline{AB}$

What is the fundamental rule for two cells in a Karnaugh map to be considered adjacent ?

Options :

1. They must be located diagonally from one another
2. They must differ in only one variable
3. They must contain the exact same literals
4. They must differ in atleast two variables

When $A = B = 0$ in the logic circuit shown in the figure, Q_1 , Q_2 outputs will be, respectively



Options :

1. 1, 0 OR 0, 1
2. 1, 1
3. 0, 0
4. 0, 1 OR 1, 1

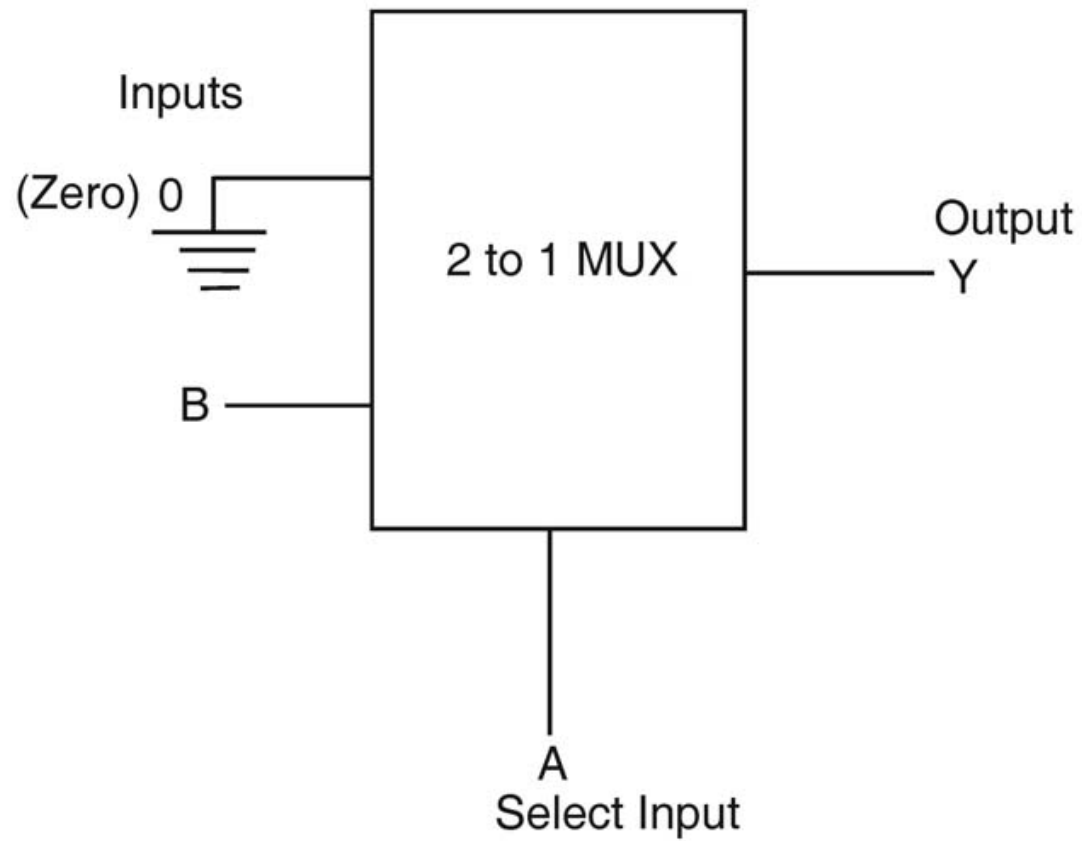
The truth table shown below is for

Inputs		Output
X	Y	P
1	0	1
0	1	1
1	1	1

Options :

1. NAND gate
2. NOR gate
3. OR gate
4. Exclusive OR gate

The output Y of the multiplexer circuit shown in figure is



Options :

1. 0
2. B
3. AB

4. $A + B$

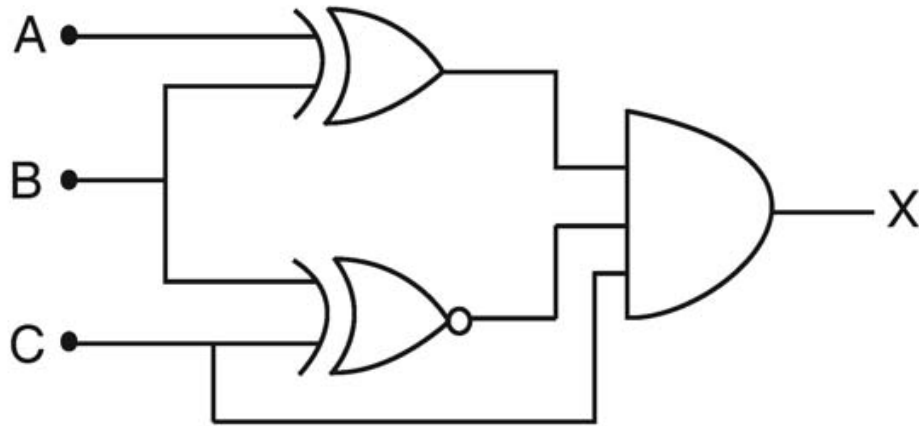
Question Number : 58 Question Id : 464662158 Question Type : MCQ Option Shuffling : No

A master-slave J – K flip-flop with $J = 1$ and $K = 1$ has a 20 kHz clock input. The Q output is

Options :

1. Constantly high
2. A 10 kHz square wave
3. Constantly low
4. A 20 kHz square wave

In the following figure input conditions are shown which produces $X = 1$, then input values are



Options :

1. $A = 1, B = 1, C = 1$
2. $A = 1, B = 1, C = 0$
3. $A = 1, B = 0, C = 0$
4. $A = 0, B = 1, C = 1$

Setup time in a flip-flop is defined as,

Options :

1. Time for output to change after clock edge
2. Minimum time during which data must be stable after the clock edge
3. Minimum time during which data must be stable before (prior) the active clock edge
4. Time taken for data to propagate to the output

In which addressing mode, the operand is actually present in the instruction ?

Options :

1. Direct mode
2. Indirect mode
3. Immediate mode
4. Register mode

What is the primary function of the control address register in a microprogrammed control unit ?

Options :

1. It holds microinstruction
2. It issues 'Read' command to the memory
3. It contains the address of the next microinstruction
4. It stress the status of the ALU flags

Question Number : 63 Question Id : 464662163 Question Type : MCQ Option Shuffling : No

What is the total capacity of 16K×4 DRAM ?

Options :

1. 16 KB
2. 32 KB
3. 64 KB
4. 8 KB

The portion of the root locus corresponding to $0 \leq k < \infty$ is called

Options :

1. Complementary root locus
2. Complete root locus
3. Root contours
4. Root locus

Question Number : 65 Question Id : 464662165 Question Type : MCQ Option Shuffling : No

If the damping ratio is 0.5 then the value of resonant peak is

Options :

1. 2.308
2. 1.154
3. 1.01
4. None of the above

Question Number : 66 Question Id : 464662166 Question Type : MCQ Option Shuffling : No

A compensator has transfer function $G(s) = \frac{s+5}{s+50}$. It is a

Options :

1. lag compensator
2. lead compensator
3. lead lag compensator
4. pure gain compensator

Question Number : 67 Question Id : 464662167 Question Type : MCQ Option Shuffling : No

Characteristics equation is given $s^3 + 6s^2 + 11s + K = 0$. For which value of K, the system is stable ?

Options :

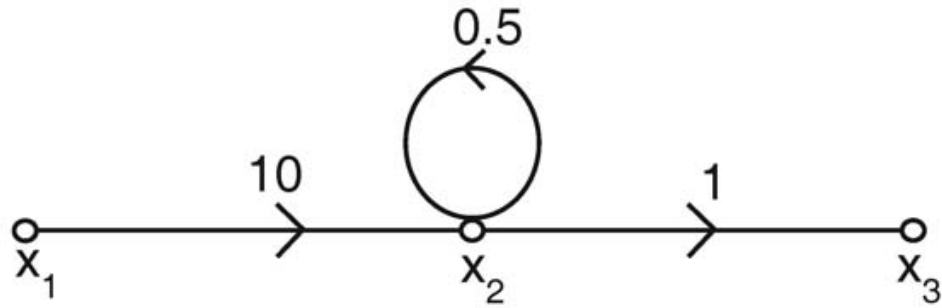
1. $K > 0$

2. $0 < K < 66$

3. $K < 30$

4. No positive K is stable

For the following signal flow graph x_3/x_1 is



Options :

1. 20
2. 10
3. 1
4. 0.1

A system has 8 poles and 3 zeros. The slope of its highest frequency asymptote in its magnitude plot is

Options :

1. +150 dB/decade
2. +100 dB/decade
3. -100 dB/decade
4. -150 dB/decade

A linear system at rest is subject to an input signal $r(t) = 1 - e^{(-t)}$.

The response of the system for $t > 0$ is given by $c(t) = 1 - e^{(-2t)}$.

The transfer function of the system is

Options :

1. $\frac{(s+2)}{(s+1)}$

2. $\frac{(s+1)}{(s+2)}$

3. $\frac{2(s+1)}{(s+2)}$

4. $\frac{(s+1)}{2(s+2)}$

Question Number : 71 Question Id : 464662171 Question Type : MCQ Option Shuffling : No

If the number of poles is “m” and the number of zeros is “n” to plot a root-locus, then the number of root-locus segments going to infinity is

Options :

1. $m + n$
2. $m - n$
3. m/n
4. mn

Which of the following system is type 1 ?

Options :

$$G(s) = \frac{(s+1)(s+2)}{s(s+4)(s+3)}$$

1.

$$G(s) = \frac{(s+4)(s+2)}{s^2(s+1)(s+3)}$$

2.

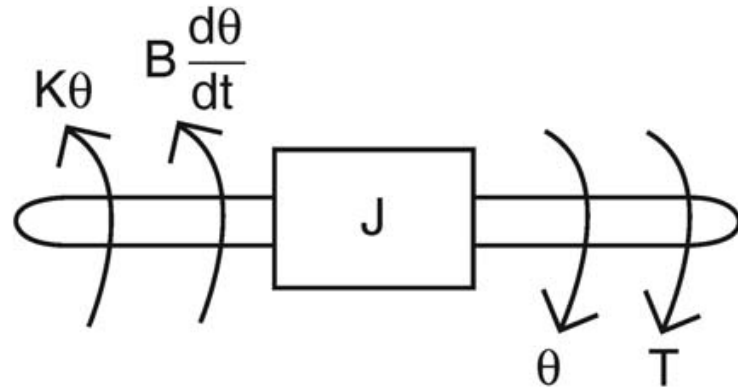
$$G(s) = \frac{(s+2)(s+5)}{s^3(s+1)(s+6)}$$

3.

$$G(s) = \frac{(s-2)(s+6)}{(s+7)(s-4)}$$

4.

Write the transfer function from the following free body diagram.



Options :

1.
$$G(s) = \frac{1}{Js + Bs^2 + K}$$

2.
$$G(s) = \frac{1}{Js^2 + Bs + K}$$

3.
$$G(s) = \frac{s}{Js^2 + Bs + K}$$

4.
$$G(s) = \frac{s+1}{Js^2 + Bs + K}$$

Question Number : 74 Question Id : 464662174 Question Type : MCQ Option Shuffling : No

The stability condition of a closed loop system depends primarily on

Options :

1. Only the magnitude of GH
2. Only phase of GH
3. Only the sign of GH
4. Both magnitude and phase of GH

Bode plots are often called 'asymptotic plots' because

Options :

1. They use logarithmic scales
2. They are exact plots
3. They show only magnitudes
4. They are drawn using straight-line approximations

What effect does phase lead compensation have on the steady state error of the system ?

Options :

1. It is significantly reduced
2. It is completely eliminated
3. It is doubled
4. It is not affected

Determine the peak frequency deviation (Δf) and modulation index (m) for an FM modulator with a deviation sensitivity $K_1 = 5 \text{ kHz/V}$ and a modulating signal

$$V_m(t) = 2 \cos(2\pi 2000t)$$

- a. $\Delta f = 400 \text{ Hz}$, $m = 5$
- b. $\Delta f = 10 \text{ kHz}$, $m = 0.2$
- c. $\Delta f = 10 \text{ kHz}$, $m = 5$
- d. $\Delta f = 400 \text{ Hz}$, $m = 0.2$

Which of these are correct ?

Options :

- 1. c only
- 2. b only
- 3. a and c only
- 4. d only

What is the unmodulated carrier power ' P_c ' and total power developed ' P_t ' by the AM wave in a load of 100Ω when the peak voltage of the carrier is 100 volt and the modulation index is 0.4 ?

Options :

1. $P_c = 50 \text{ W } P_t = 54 \text{ W}$

2. $P_c = 500 \text{ W } P_t = 540 \text{ W}$

3. $P_c = 100 \text{ W } P_t = 108 \text{ W}$

4. $P_c = 50 \text{ W } P_t = 108 \text{ W}$

Question Number : 79 Question Id : 464662179 Question Type : MCQ Option Shuffling : No

Assume on MSK modem with $f_m = 1200$ and $f_s = 1800$ Hz. The bit rate is 1200 bps.
The modulation index is

Options :

1. $m = 1$
2. $m = 0.5$
3. $m = 0$
4. $m = -1$

Question Number : 80 Question Id : 464662180 Question Type : MCQ Option Shuffling : No

DPCM reduces

Options :

1. Quantization error
2. Transmission bandwidth
3. Both (1) and (2)
4. Modulation index

The entropy of an information source is a measure of

Options :

1. Information transmission rate
2. Average information content per source symbol
3. Channel noise
4. Signal bandwidth

If the minimum Hamming distance of a Hamming code is 3, it is capable of detecting and correcting a maximum of _____ and _____ errors, respectively.

Options :

1. 1, 2

2. 2, 1

3. 2, 2

4. 1, 1

In channel coding, $U = 1010$, $V = 0111$ and $W = 1001$ are the codewords. The hamming distance between U and V, hamming distance between U and W and hamming distance between V and W respectively is

Options :

1. 1, 2, 1
2. 2, 1, 2
3. 3, 2, 3
4. 2, 3, 2

In an AM receiver, following are the desirable characteristics of RF amplifier.

- a. Low thermal noise and low noise figure
- b. Low intermodulation and harmonic distortion
- c. High image frequency rejection ratio and moderate selectivity
- d. High thermal noise and low noise figure

Which of these statements is/are correct ?

Options :

- 1. a only
- 2. c and d only
- 3. a, b and d
- 4. a, b and c

Question Number : 85 Question Id : 464662185 Question Type : MCQ Option Shuffling : No

Cyclic Redundancy Check (CRC) is primarily used for

Options :

1. Error correction
2. Source coding
3. Error detection using polymeric division
4. Data compression

In view with channel coding

- a. In block codes, each block of k message bits is encoded into a block of n bits. The $(n-k)$ are check bits.
- b. Message bits appear at the beginning of the code word in systematic block code.
- c. The message bit stream is encoded in a continuous fashion, rather than in piecemeal as in block codes, in convolution codes.
- d. Cyclic codes are a subclass of linear block codes.

Which of these statements is/are correct ?

Options :

- 1. a only
- 2. a and b only
- 3. a, b, c and d
- 4. a, b and d only

Question Number : 87 Question Id : 464662187 Question Type : MCQ Option Shuffling : No

When the modulating frequency is doubled, the modulation index is halved and the modulating voltage remains constant. This modulation system is

Options :

1. Amplitude modulation
2. Phase modulation
3. Frequency modulation
4. All of the above

One of the following types of noise is important at high frequencies. It is the

Options :

1. Shot noise
2. Random noise
3. Impulse noise
4. Transit time noise

In a PCM system, the two major sources of error are

Options :

1. Sampling noise and aliasing
2. Quantization noise and channel noise
3. Thermal noise and intersymbol interferences
4. Phase noise and frequency offset

For total internal reflection inside the fiber, what must be the relationship between the angle of incidence (θ_i) and the critical angle of incidence (θ_c) ?

Options :

1. $\theta_i > \theta_c$

2. $\theta_i < \theta_c$

3. $\theta_i = \theta_c$

4. $\theta_i = 0^\circ$

The Smith chart is primarily a graphical representation of

Options :

1. Voltage and current on transmission line
2. Complex power and reflected power
3. Reflection coefficient and normalised impedance
4. Time-domain signal behaviour

The Poynting vector represents

- a. Electric flux density
- b. Magnetic flux density
- c. Power flow per unit area
- d. Charge density

Options :

- 1. a only
- 2. c only
- 3. b only
- 4. d only

The phase velocity of waves propagating in a hollow metal waveguide is

Options :

1. Equal to the group velocity
2. Equal to the velocity of light in free space
3. Less than the velocity of light in free space
4. Greater than the velocity of light in free space

Skin depth decreases with

- a. Increasing frequency
- b. Decreasing frequency
- c. Increasing conductivity

Options :

- 1. a only
- 2. Both b and c
- 3. b only
- 4. Both a and c

Given that the frequency of a plane electromagnetic wave increases four times, what will be the effect on the skin depth of a wave which is incident normally on a good conductor ?

Options :

1. It increases by a factor of 2
2. It decreases by a factor of 2
3. It remains the same
4. It increases by a factor of 4

Question Number : 96 Question Id : 464662196 Question Type : MCQ Option Shuffling : No

Electromagnetic waves are transverse in nature due to

Options :

1. Reflection
2. Diffraction
3. Polarization
4. Interference

Question Number : 97 Question Id : 464662197 Question Type : MCQ Option Shuffling : No

A broadside array has its maximum radiation

Options :

1. Along the axis of the array
2. Perpendicular to the axis of the array
3. At an angle of 45° to the axis of the array
4. At an angle of 60° to the axis of the array

Match the transmission line concept with its description

- | | |
|-----------------------------|--|
| a. Characteristic impedance | i. Graphical tool for impedance matching |
| b. Smith chart | ii. Ratio of voltage to current of a travelling wave |
| c. S-parameters | iii. Describes power wave relations at ports |
| d. Quarter wave parameter | iv. Used for impedance transformation |

Options :

1. a – i, b – ii, c – iv, d – iii
2. a – ii, b – iv, c – i, d – iii
3. a – i, b – iv, c – iii, d – ii
4. a – ii, b – i, c – iii, d – iv

If A, B, C and D are the constants of the medium transmission line, which of the following relation is correct ?

Options :

1. $AB - CD = -1$

2. $AD + BC = 1$

3. $BC - AD = -1$

4. $AC - BD = 1$

Question Number : 100 Question Id : 464662200 Question Type : MCQ Option Shuffling : No

The input impedance of a very small dipole antenna is

Options :

1. Capacitive
2. Resistive
3. Inductive
4. Capacitive or inductive based on the input frequency